Metals and Non-metals

The big idea

Elements can be sorted as **metals** (iron, copper, sodium…) and **non-metals** (carbon, sulphur, oxygen…). Their properties, and how readily they react, decide where we use them and how we extract them from the earth.

Physical properties

Property	Metals	Non-metals
Lustre	Shiny	Dull (except iodine)
Malleable / ductile	Yes (gold, silver most)	No, brittle solids
Conduct heat and electricity	Yes (silver and copper best)	No (except graphite)
State	Solids (except mercury)	Solids, liquid (bromine) or gases
Sonorous	Yes	No

Exceptions to remember: sodium and potassium are soft (cut with a knife); gallium and caesium melt in your palm; diamond (carbon) is the hardest natural substance.

Chemical properties

HOW METALS REACT

Metal + oxygen → metal oxide (basic). **Amphoteric** oxides like Al_2O_3 and ZnO react with both acids and bases.
Metal + water → metal hydroxide (or oxide) + hydrogen. Na and K react violently with cold water; Mg with hot water; Al, Fe, Zn only with steam; Cu, Ag, Au not at all.
Metal + dilute acid → salt + hydrogen (not Cu, Ag, Au).
A more reactive metal **displaces** a less reactive one from its salt solution.

The reactivity series

K > Na > Ca > Mg > Al > Zn > Fe > Pb > [H] > Cu > Hg > Ag > Au

Metals above hydrogen displace it from dilute acids. Non-metal oxides are **acidic** (SO_2, CO_2).

Ionic compounds

Metals lose electrons and non-metals gain them, forming ions that attract strongly. Example: Na (2,8,1) gives one electron to Cl (2,8,7) to form Na^+Cl^-.

PROPERTIES OF IONIC COMPOUNDS

Hard solids with **high melting and boiling points**, usually soluble in water but not in kerosene or petrol, and they **conduct electricity when molten or dissolved** (ions are free to move), not as solids.

Extraction of metals (metallurgy)

1. **Concentration of ore**: remove gangue (sand, rock).
2. **Low reactivity** metals (Hg, Cu): heat the sulphide ore in air.
3. **Medium reactivity** (Zn, Fe, Pb): sulphide → oxide by **roasting**; carbonate → oxide by **calcination**; then reduce with carbon: ZnO + C → Zn + CO. Highly

reactive Al can reduce oxides in the **thermite reaction**: $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$ (used to weld railway tracks).

4. **High reactivity** (Na, Mg, Ca, Al): **electrolysis** of the molten chloride or oxide.
5. **Refining**: electrolytic refining, with impure metal as anode and pure metal as cathode.

Corrosion and its prevention

Rusting needs **both air and water**. Prevent it by painting, oiling, greasing, **galvanising** (coating with zinc) or making **alloys**: stainless steel (iron + nickel + chromium), brass (copper + zinc), bronze (copper + tin), solder (lead + tin, low melting point).

Quick recap

- Metals: lustrous, malleable, ductile, good conductors, form basic oxides.
- Reactivity series decides displacement and extraction method.
- Ionic compounds: high melting point, conduct when molten or in solution.
- Prevent corrosion by coating, galvanising and alloying.

Practice questions

1. Which metal is a liquid at room temperature?
   (a) Sodium (b) Mercury (c) Gallium (d) Bromine

2. An amphoteric oxide among these is
   (a) Na_2O (b) CaO (c) Al_2O_3 (d) SO_2

3. Which metal can displace copper from copper sulphate solution?
   (a) Zinc (b) Silver (c) Gold (d) Mercury

4. Ionic compounds conduct electricity
   (a) in the solid state (b) when molten or dissolved in water (c) never (d) only in kerosene

5. Galvanisation means coating iron with
   (a) tin (b) copper (c) zinc (d) chromium

6. Highly reactive metals like sodium are extracted by
   (a) heating with carbon (b) roasting (c) electrolysis of their molten compounds (d) heating alone

Answers: 1. b 2. c 3. a 4. b 5. c 6. c